UNIVERSITÉ LAVAL

September 30th, 2023

Subject: Application, Assistant Professor position in Molecular Ecology

Dear Dr. Declan Ali and members of the Search Committee,

I am pleased to submit my application for the recently advertised Assistant Professor position in Molecular Ecology at the University of Alberta. My research program in molecular evolution is an excellent match to the description of the ideal candidate. I explore how genetic variation and phenotypic plasticity contribute to acclimation, adaptation, phenotypic divergence, and speciation in natural and controlled systems. I incorporate population genomics and quantitative genetic approaches into the analysis of complex multi-omic (genomic, transcriptomic, epigenomic, and recently proteomic) datasets to better understand different sources of molecular variation, interactions among them, and the resultant adaptive (or maladaptive) evolutionary implications for populations. My research furthers our understanding of how organisms acclimate to changing environments, how molecular variation can contribute to stable phenotypic variation, and how different sources of molecular variation interact and lead to evolutionary change. I have also worked on collaborative projects that focused on population genetics in conservation and speciation which align with the strengths of the Department of Biological Sciences in applied ecology and conservation. My research program would foster cross-disciplinary collaborations, applying multi-omic tools to both basic evolutionary questions and applied conservation issues. I have been developing this research program as an NSERC Postdoctoral Fellow in the lab of Louis Bernatchez at Université Laval and during my PhD in Daniel Heath's lab at the University of Windsor.

My goal is to establish an inclusive, productive, and innovative interdisciplinary research group in the Department of Biological Sciences at the University of Alberta. My work would complement ongoing research on genomics, physiology, and plasticity in the department and other departments across campus, utilizing the exceptional infrastructure such as the Molecular Biology Services Unit, Biotron aquatics facility, and Bamfield Marine Sciences Centre. The potential to collaborate with more than 50 colleagues in the department with diverse research programs would allow me to develop novel, interdisciplinary collaborations due to the flexibility of the genomic approaches I use, the ease of applying them to diverse taxa, and their possible applications at the intersection between evolutionary, molecular, and conservation biology. My epigenetic and multi-omic research lies at the boundary between physiology and evolutionary genetics and will serve as a bridge between different research programs in the department. For example, my expertise could easily be applied to ongoing research on fish physiology by Drs. Tamzin Blewett, Greg Goss, and Keith Tierney to study the contributions of molecular plasticity towards physiological responses. My work would also complement research on environmental responses and phenotypic plasticity by Drs. Janice Cooke, Maya Evenden, Stephanie Green, Zach Hall, George Owttrim, and Olav Rueppell. DNA methylation also contributes towards sex determination and developmental plasticity which would complement the research of Drs. Kimberley Mathot and David Pilgrim, though I see potential for many more collaborations using my multi-omic approaches within the department. Interdisciplinary, collaborative, and open science offers the best opportunity to understand how populations and species evolve and cope with environmental change. This sort of collaborative research presents an opportunity to determine how species will cope with climate change and other anthropogenic effects leading to effective conservation efforts. It also allows us to unravel the complexities of adaptation and evolution, answering fundamental and timely questions in evolutionary biology.

I currently have 13 papers that are published or in press at scholarly journals and three articles in review. I have published empirical research in journals such as Molecular Biology and Evolution (IF=8.8, 23 citations since 2020), Molecular Ecology (IF=6.6, 12 citations since 2022 and 31 citations since 2016), Proceedings of the Royal Society B: Biological Sciences (IF=5.5, 10 citations since 2022), Heredity (IF=3.8, 26 citations since 2021), Molecular Ecology Resources (IF=8.7, 1 citation since 2023), and Evolution (IF=3.3, 1 citation since 2022). I share first authorship on highly collaborative reviews in Nature Reviews Genetics (IF=59.9, in press) and Trends in Ecology and Evolution (IF=20.6, 60 citations since 2021). I have an h-index of 8 with less than three years of postdoctoral experience, reinforcing the importance and timeliness of my research program. Beyond journal and publication statistics, my research has provided innovative insights into the role of epigenomic and multi-omic variation in evolutionary biology.

I have considerable experience teaching undergraduate courses and mentoring graduate and undergraduate students. I have mentored [redacted]. I have taught classes and labs in courses including Evolution, Genetics, and Introductory Molecular Biology. I would be comfortable teaching courses including, but not limited to, Foundations of Molecular Genetics, Genetics of Eukaryotic Chromosomes, Mechanisms of Evolution, and Genetic Analysis of Populations. I have had the pleasure of working with students from diverse backgrounds and perspectives that have brought science to life in the classroom and lab. Topics in genetics and molecular biology can be difficult to understand but I excel at simplifying complex topics. I realize that every student is different and will require different levels of support and learning approaches. I approach teaching, mentorship, and all collaborations with the view that science should be a collaborative and mutualistic process where all parties benefit. This has helped me develop excellent communication skills and teaching approaches while remaining sensitive to the needs of individual students. I believe that I have as much to learn from my students as they can learn from me. This includes identifying, minimizing, and dismantling barriers to success that may prevent underrepresented group from pursuing education and careers in science. Every new mentorship and teaching opportunity has taught me something new about students' diverse needs, barriers to their success, and the importance of considering an individual student's needs and intersectionality while helping them reach their maximum potential. This openness extends beyond science itself for me; for example, I have been teaching myself French while living in Québec in order to participate in formal and informal discussions with my francophone mentees and colleagues, valorising our Canadian bilingual culture. I learn with every experience and find new ways to approach teaching and mentorship to instill my students with the passion for science and comprehensive understanding needed for them to excel in their studies and future careers.

I am a Canadian citizen [redacted]. I have included my curriculum vitae and representative publications in addition to my statements on research, teaching, and my commitment to equity, diversity, and inclusion. Drs. Louis Bernatchez, Daniel Heath, and Julie Smit will provide letters of recommendation upon request. I believe I am the ideal candidate for this position due to my innovative, rigorous, collaborative research program. My use of multi-omic data to answer broad evolutionary and applied questions in molecular ecology will enhance ongoing research in the Department of Biological Sciences, uniting genomic and ecological research due to the unique placement of my research interests. Thank you for considering my candidature for this position.

Sincerely,

Dr. Clare Venney

CLARE VENNEY

Current affiliation **NSERC Postdoctoral Fellow, Bernatchez Lab**
Institut de biologie intégrative et des systèmes
Université Laval
Québec, QC, Canada
Citizenship Canadian
E-mail clarevenney@gmail.com
clare.venney.1@ulaval.ca
Phone [redacted]
Publication record Google Scholar

CAREER ASPIRATIONS

I am a molecular ecologist interested in the evolutionary implications of different sources of molecular variation for natural populations. I am particularly interested in (1) how genomic, epigenomic, and transcriptomic variation can lead to phenotypic responses to short-term environmental challenges, (2) how molecular variation contributes to phenotypic differentiation, and (3) how molecular plasticity can lead to genetic differentiation and adaptation. I aim to lead a primary research laboratory at a Canadian university to explore these questions.

EDUCATION AND RESEARCH EXPERIENCE

NSERC Postdoctoral Fellow (2023-present), **Postdoctoral Researcher** (2020-2023) Institut de biologie intégrative et des systems, Université Laval. **Research focus:** DNA methylation in phenotypic plasticity, epigenetic inheritance, transcriptional and phenotypic variation, interactions with genetic variation, induced mutagenesis, and speciation. Supervised by Dr. Louis Bernatchez.

PhD (2014-2020) Environmental Science, Great Lakes Institute for Environmental Research, University of Windsor. **Thesis title:** Epigenetic contributions to early life history variation in Chinook salmon. Supervised by Dr. Daniel Heath.

BSc (2010-2014) Honours Biological Sciences with Thesis, minor in Biochemistry, University of Windsor. **Thesis title:** Contaminant exposure and its epigenetic effects on stress response: testing for environmentally-induced methylation changes in brown bullhead (*Ameiurus nebulosus*). Supervised by Dr. Daniel Heath.

PEER-REVIEWED PUBLICATIONS

I have 13 published or accepted papers with approximately 188 citations to date (average=15.4 citations per paper), and three manuscripts in progress. My h-index is currently 8. My five most cited papers were published in Trends in Ecology and Evolution (IF=20.6, 60 citations), Molecular Ecology (IF=6.6, 12 and 31 citations each), Heredity (IF=3.8, 26 citations), and Molecular Biology and Evolution (IF=8.8, 23 citations).

***Equal contribution is denoted by an asterisk.**

1) **Venney CJ**, Anastasiadi D, Wellenreuther M, Bernatchez L. (*In Review*) The evolutionary complexities of DNA methylation: from plasticity to genetic evolution. *Genome Biology and Evolution* (GBE-230826).
2) Sadeghi J, **Venney CJ**, Wright S, Watkins J, Manning D, Bai E, Frank C, Heath DD. (*In Review*) Aquatic bacterial community connectivity: the effect of hydrological flow on community diversity and composition. *Environmental DNA* (Submitted August 21st, 2023; EDN3-2023-0124).
3) **Venney CJ**, Mérot C, Normandeau E, Rougeux C, Laporte M, Bernatchez L. (*In Revision*) Epigenetic differentiation between *Coregonus* species pairs: potential for epigenetically induced mutation to contribute to speciation. *Genome Biology and Evolution* (Submitted July 6th, 2023; GBE-230707).
4) Bernatchez L*, Ferchaud AL*, Berger CS*, **Venney CJ***, Xuereb A*. (*Accepted*) Genomics for monitoring and understanding species response to global climate change. *Nature Reviews Genetics* (NRG-22-076V2).
5) **Venney CJ**, Bouchard R, April J, Normandeau E, Lecomte L, Côté G, Bernatchez L. (2023) Captive rearing effects on the methylome of Atlantic salmon after oceanic migration: sex-specificity and intergenerational stability. *Molecular Ecology Resources*.
6) **Venney CJ***, Cayuela H*, Rougeux C, Laporte M, Mérot C, Normandeau E, Leitwein M, Dorant Y, Præbel K, Kenchington E, Clément M, Sirois P, Bernatchez L. (2022) Genome-wide DNA methylation predicts environmentally-driven life history variation in a marine fish. *Evolution* 77(1): 186-198.
7) **Venney CJ**, Wellband KW, Normandeau E, Houle C, Garant D, Audet C, Bernatchez L. (2022) Thermal regime during parental sexual maturation, but not during offspring rearing, modulates DNA methylation in brook charr (*Salvelinus fontinalis*). *Proceedings of the Royal Society B: Biological Sciences* 289(1974): 20220670.
8) Mérot C, Stenløkk K, **Venney CJ**, Laporte M, Moser M, Normandeau E, Árnyasi M, Kent M, Rougeux C, Flynn J, Lien S, Bernatchez L. (2022) Genome assembly, structural variants, and genetic differentiation between lake whitefish young species pairs (*Coregonus* sp.) with long and short reads. *Molecular Ecology* 32: 1458-1477.
9) Walter RP, **Venney CJ**, Mandrak NE, Heath DD. (2022) Conservation implications of revised genetic structure resulting from new population discovery: the threatened eastern sand darter (*Ammocrypta pellucida*) in Canada. *Journal of Fish Biology* 100: 92-98.
10) Anastasiadi D*, **Venney CJ***, Bernatchez L, Wellenreuther M. (2021) Epigenetic inheritance and reproductive mode in plants and animals. *Trends in Ecology and Evolution* 36(12): 1124-1140.
11) **Venney CJ**, Sutherland BJG, Beacham TD, Heath DD. (2021) Population differences in Chinook salmon (*Oncorhynchus tshawytscha*) DNA methylation: genetic drift and environmental factors. *Ecology and Evolution* 11(11): 6846-6861.
12) Roy D, Lehnert S, **Venney CJ**, Walter R, Heath DD. (2021) NGS-µsat: bioinformatics framework supporting high throughput microsatellite genotyping from next generation sequencing platforms. *Conservation Genetics Resources* 13: 161-173.
13) **Venney CJ**, Wellband KW, Heath DD. (2021) Rearing environment affects the genetic architecture and plasticity of DNA methylation in Chinook salmon. *Heredity* 126: 38-49.
14) **Venney CJ**, Love OP, Drown EJ, Heath DD. (2020) DNA methylation profiles suggest intergenerational transfer of maternal effects. *Molecular Biology and Evolution* 37(2): 540-548.
15) Semeniuk CAD, Capelle PM, Dender MGE, Devlin R, Dixon B, Drown J, Heath J, Hepburn R, Higgs DM, Janisse K, Lehnert S, Love OP, Mayrand J, Mickle M, Pitcher TE, Neff B, Semple SL, Smith JL, Toews S, **Venney CJ**, Wellband K, Heath DD. (2019) Domestic-wild hybridization to improve aquaculture performance in Chinook salmon. *Aquaculture* 511: 734255.
16) **Venney CJ**, Johansson ML, Heath DD. (2016) Inbreeding effects on gene-specific DNA methylation among tissues of Chinook salmon. *Molecular Ecology* 25(18): 4521-4533.

CONTRIBUTED ORAL AND POSTER PRESENTATIONS

1) **Venney CJ**, Mérot C, Normandeau E, Rougeux C, Laporte M, Bernatchez L. (2023) Epigenetic differentiation between benthic-limnetic whitefish species pairs: a potential source of polymorphism and genetic divergence. Coregonid Symposium 2023, September 25-29, Évian-les-Bains, France.
2) **Venney CJ**. (2023) The role of DNA methylation in evolution: from short-term plasticity to genetic evolution. Invited Seminar, Bielefeld University, April 25, Online.
3) **Venney CJ**, Cayuela H, Rougeux C, Laporte M, Mérot C, Normandeau E, Leitwein M, Dorant Y, Præbel K, Kenchington E, Clément M, Sirois P, Bernatchez L. (2023) DNA methylation underlies intraspecific differences in reproductive tactics in a marine fish. Society of Molecular Biology and Evolution (SMBE) Everywhere GS9: Epigenomics and Evolution, February 28, Online.
4) **Venney CJ**, Mérot C, Wellband KW, Normandeau E, Rougeux C, Laporte M, Bernatchez L. (2023) DNA methylation in whitefish speciation (*Coregonus* sp.): mutagenesis, genetic assimilation, and transcriptional effects. Gordon Research Conference on Speciation, January 29-February 3, Barga, Italy.
5) **Venney CJ**, Wellband KW, Normandeau E, Houle C, Garant D, Audet C, Bernatchez L. (2022) L’environnement thermique des parents détermine l’état de la méthylation d’ADN des rejetons chez l’omble de fontaine (*Salvelinus fontinalis*). Ressources Aquatiques Québec Annual Meeting, November 7-9, Québec, QC.
6) **Venney CJ**, Wellband KW, Normandeau E, Houle C, Garant D, Audet C, Bernatchez L. (2022) Parental, but not juvenile, temperature modulates offspring DNA methylation in brook charr. American Fisheries Society (AFS) Annual Meeting, August 21-25, Spokane, WA.
7) Laporte M, Leitwein M, Mérot C, **Venney CJ**, Cayuela H, Flynn J, Dion-Côté A-M, Normandeau E, Bernatchez L. (2021) Global DNA hypomethylation in malformed lake whitefish backcross: a post-zygotic barrier in nascent species. Evolution, June 25-29, Online.
8) Mérot C, Rougeux C, Stenløkk KSR, Normandeau E, **Venney CJ**, Wellband K, Lien S, Bernatchez L. (2021) The genetic architecture of ecological speciation in whitefish (*Coregonus* sp.): assessing the dual role of point mutations and structural variants. Evolution, June 25-29, Online.
9) **Venney CJ**, Wellband KW, Heath DD. (2021) Hatchery vs. semi-natural rearing influences the genetic architecture of DNA methylation in Chinook salmon. Canadian Conference for Fisheries Research (CCFFR), February 15-19, Online.
10) **Venney CJ**, Love OPL, Drown EJ, Heath DD. (2020) Intergenerational transfer of maternal effects via DNA methylation in Chinook salmon. Canadian Conference for Fisheries Research (CCFFR), January 2-5, Halifax, NS.
11) Roy D, Lehnert S, **Venney CJ**, Walter RP, Heath DD. (2019) NGS-µsat: bioinformatics framework supporting high throughput microsatellite genotyping from next generation sequencing platforms. Canadian Society for Ecology and Evolution (CSEE), August 18-21, Fredericton, NB.
12) Mychek-Londer JG, **Venney CJ**, Heath DD (2019). Metabarcoding of native and invasive prey in stomach content DNA (scDNA) of commercially harvested Lake Erie fishes. Canadian Conference for Fisheries Research (CCFFR), January 2-6, London, ON.
13) **Venney CJ,** Wellband KW, Heath DD (2018). Additive, non-additive, and maternal effects on DNA methylation among different salmon rearing environments. Canadian Conference for Fisheries Research (CCFFR), January 4-7, Edmonton, AB.
14) **Venney CJ,** Roy D, Heath DD (2017). DNA methylation as an epigenetic mechanism for local adaptation in Chinook salmon. World Recreational Fishing Conference, July 16-20, Victoria, BC.
15) **Venney CJ,** Wellband KW, Heath DD (2017). Gene-specific parental influences on DNA methylation as a source of intraspecific variation. Canadian Conference for Fisheries Research (CCFFR), January 5-8, Montreal, QC.
16) **Venney CJ**, Johansson ML, Heath DD (2016). Ontogenetic and tissue-specific DNA methylation in Chinook salmon. Canadian Conference for Fisheries Research (CCFFR), January 7-9, St John’s, NL.
17) **Venney CJ**, Heath DD (2015). Population-level epigenetic responses to stressors in aquaculture.

Development of high performance Chinook salmon stocks for commercial aquaculture: Strategic Grant Partners Meeting, June 4, Campbell River, BC.

18) Walter RP, **Venney CJ**, Gnyra E, Soderberg LI, Heath DD (2013). Massive reticulate evolution in Bullheads (*Ameiurus*) from the Lower Great Lakes. Canadian Conference for Fisheries Research (CCFFR), January 4, Windsor, ON.

BOOK CHAPTERS

Doci F, **Venney CJ**, Spencer C, Diemer K. "Epigenetics and Law: The Quest for Justice." In Epigenetics in Society, edited by Michael Crawford, p. 257-278. Windsor, ON: Emerging Scholars Press, October 2015.

GREY LITERATURE

Venney CJ, Audet C, Garant D, Bernatchez L. (2023) Production d'omble de fontaine adaptée aux changements climatiques: évaluation de l'importance des effets épigénétiques. OURANOS (Link; in French only)

NONTRADITIONAL PUBLICATIONS

Halverson A (host), **Venney CJ** (invitee), Bouchard R (invitee). (December 18, 2022). How hatcheries affect salmon epigenetics, with Clare Venney and Raphaël Bouchard [Audio Podcast Episode]. In *The Fisheries Podcast*. (Link)

SCHOLARSHIPS AND AWARDS

2023-2025	NSERC Postdoctoral Fellowship (**$90,000**)
2022	Ressources Aquatiques Québec (RAQ) Bonus for Scientific Publication (**$500**)
2022	Ressources Aquatiques Québec (RAQ) Travel Award (**$500**)
2021	GLIER Lum Clark Award for Research Excellence (**$500**)
2017	Ontario Graduate Scholarship (**$15,000**)
2017	Faculty of Science Student Travel Fund (**$500**)
2017	Canadian Conference for Fisheries Research Clemens-Rigler Travel Award (**$225**)
2016-2020	University of Windsor PhD Entrance Scholarship (**$24,000**)
2016	Canadian Conference for Fisheries Research Clemens-Rigler Travel Award (**$450**)

MENTORSHIP AND TRAINING EXPERIENCE

Graduate Students

[redacted] [redacted]

[redacted]

Undergraduate Students

[redacted]

Postdoctoral Researchers

[redacted]

UNDERGRADUATE TEACHING EXPERIENCE

Evolution (BIOL-3142: Winter 2020, Winter 2017): Gave guest lectures for ~180 students, directed class discussions, marked assignments and exams.

Genetics (BIOL-2111: Fall 2019, Fall 2018, Fall 2017, Fall 2016, Fall 2015, Fall 2014): Directed laboratories, marked weekly assignments, wrote and marked the final lab exam.

Special Topics: Evolutionary, Ecological, and Environmental Genetics (BIOL-4008: Winter 2019): Co-instructed the course. Designed and taught lessons on the use of GenAlEx, R, and other software to teach undergraduate students to handle real genetic data. Assisted the class with designing and performing an experiment on microbial connectivity which resulted in the submission of a manuscript to a peer-reviewed journal.

Introductory Molecular Biology (BIOL-2033: Winter 2018, Winter 2016, Winter 2015): Directed laboratories, marked assignments and exams.

Selected anonymous feedback from Genetics students (Average instructor score: 4.55/5)

- "Instructor was amazing. Really loved her! Explains concepts clearly and when asked questions they are properly answered."
- "Great lab instructor. She was very clear when explaining concepts and answering questions."
- "Probably the best one I've had in university – keep up the great job!"
- "This lab made me like genetics. I wish we could have more genetics labs."
- "Clare was excellent. She demonstrated outstanding abilities in delivering material."
- "Very helpful, one of the best lab instructors I've had."

ACADEMIC SERVICE

Community Service

2021-2023 Canadian Aquatic Resources Section of the American Fisheries Society (CARS-AFS) Peter A. Larkin PhD Award Review Panel
2023 Support Our Science Nationwide Walkout, Head Local Organizer, Québec City
2022 Moderator, EcoEvo.social Mastodon Server
2018-present Peer reviewed scientific articles in Nature Ecology and Evolution, Molecular Biology and Evolution, Molecular Ecology, BMC Biology, Evolutionary Applications, Genomics, Canadian Journal of Fisheries and Aquatic Sciences, Environmental Biology of Fishes, Conservation Physiology, etc.

Committees

2016-2020 Great Lakes Institute for Environmental Research Council, University of Windsor
2016-2017 Faculty of Science Council, University of Windsor
2015-2016 Tenure Track Assistant Professor Search Committee, Great Lakes Institute for Environmental Research and The Department of Chemistry and Biochemistry, University of Windsor

Conference Organization

2022-2023 SMBEeverywhere Session Organizer, GS9: Epigenomics in Evolution
2015, 2016 GLIER Multidisciplinary Graduate Student Symposium Co-organizer

Scientific Outreach

2017, 2018 University of Windsor Science Academy, DNA Forensic Lab Leader
2017 Great Lakes Citizen Science Water Sampling Event, Volunteer Coordinator
2016 University of Windsor Science Academy, DNA Forensic Lab Assistant

Research Statement

The advent of novel genomic tools allows us to address fundamental questions in evolutionary biology, many of which have broad implications for our understanding of species responses to climate change. My research characterizes molecular drivers of phenotypic plasticity and their implications for acclimation, adaptation, and speciation. I primarily study DNA methylation – an epigenetic, covalent DNA modification that influences transcription, alternative splicing, and mutagenesis with implications for evolution, climate change adaptation, and human health, among others. Using a combination of field and experimental studies, I combine epigenomics with genomic, transcriptomic, environmental, and phenotypic data to determine the evolutionary implications of multi-omic variation in fishes. While I mostly study fishes, these methods are broadly applicable to most species. I am particularly interested in the potential for methylation to act on an evolutionary continuum ranging from rapid, transient acclimation to induced mutagenesis, genetic divergence, and evolution. My research program addresses molecular drivers of phenotypic plasticity and their contributions to evolution. Specifically, I study (1) rapid acclimation through plasticity, (2) contributions of multi-omic variation to stable phenotypic or adaptive variation, and (3) long-term genetic and evolutionary implications of DNA methylation.

Past and Current Work: I studied DNA methylation in Chinook salmon (*Oncorhynchus tshawytscha*) during my PhD, employing both quantitative and population genetic approaches to understand the genetic and environmental determinants of methylation state. I designed candidate gene methylation assays to use with large quantitative breeding designs and population analyses. I assessed the tissue specificity of DNA methylation[1], its underlying genetic architecture[2] in different environments[3] (i.e., additive/heritable, non-additive, and maternal sources of variation), and how methylation correlated with genetic and environmental variation in natural populations[4]. My doctorate helped to establish a ground-up understanding of the sources of epigenetic variation and raised new questions on the ecological and evolutionary implications of molecular plasticity.

My postdoctoral work, currently funded by an NSERC fellowship, uses a cutting-edge combination of whole methylome, genome, and transcriptome sequencing to answer diverse questions on the role of DNA methylation and multi-omic variation in the acclimation and evolution of various economically and ecologically important fishes. I have published in respected journals on topics including (1) captive rearing effects on the Atlantic salmon (*Salmo salar*) methylome during supplementation efforts and their implications for wild-born progeny,[5] (2) the effects of thermal regime on brook charr (*Salvelinus fontinalis*) methylation and the potential for epigenetic inheritance to prepare offspring for warming climates,[6] and (3) epigenetic and genetic differences driving capelin (*Mallotus villosus*) life history and spawning tactics.[7] My ongoing projects involve (1) the contributions of genetic and epigenetic variation to speciation in whitefish (*Corgeonus* spp.) species pairs, providing support for methylation-induced mutagenesis contributing to genetic divergence between species (in revision), (2) landscape genomics and epigenomics in brook charr and lake whitefish (*Coregonus clupeaformis*) in the Canadian Arctic to assess epigenetic variation among populations and its relationship with genetic variation, and (3) the effects of methylation on differential expression and alternative splicing in whitefish species pairs. Together, my research shows the importance of multi-omic variation in driving interindividual variation, acclimation, phenotypic and population differentiation, and speciation.

Future Research Directions: My research group will study the role of multi-omic variation in ecology and evolution. Little is known about the role of epigenetics in evolutionary processes compared to our knowledge on genetic variation. However, it is important to quantify the relative implications of genetic vs. plastic sources of variation in natural populations since both can affect phenotype and fitness. This research will increase our knowledge on the processes that contribute to acclimation, adaptation, and evolution, with applications in conservation and management. My lab's research will focus on fishes such as salmonids that are of economic and/or conservation concern in Canada, including westslope cutthroat trout, bull trout, Coho salmon, and Chinook salmon in western Canada. My abilities in spoken and written French facilitate inclusive nationwide collaborations in both of Canada's official languages. This research will complement ongoing departmental research in fish and fisheries with broad, timely implications for conservation and aquaculture. While my research will focus on salmonids, the techniques I use are easily applied to diverse taxa, encouraging new collaborations within and among departments. The University of Alberta is one of the most well-equipped and productive Canadian research universities which would be a huge support for building my research program. Access to the Illumina NextSeq platform in the Molecular Biology Service Unit for all DNA and RNA sequencing, the Biotron aquatics facility, and the Bamfield Marine Sciences Centre will facilitate my work. In the first five to 10 years, my lab will address questions that have arisen during my previous research:

How do molecular differences contribute to phenotypic variation? Genomic, transcriptomic, and epigenomic variation can affect phenotype, though the relative contributions of these molecular variants to physiological and phenotypic variation is understudied. My research group will characterize multi-omic variation and relate these sources of variation to fitness-related traits such as thermal resistance and stress response. We will use a combination of field studies and controlled laboratory experiments to measure phenotypes such as length, weight, condition, critical thermal maximum, hatching success, and survival to identify molecular variation associated with phenotypic variation among populations and species. Interactions between different molecular markers (e.g., genetic control over transcription or methylation state) will also be investigated to identify sources of constraint over molecular variation. Due to the potential for epigenetic inheritance contributing to heritability and parental effects, we will also study epigenomic and transcriptomic variation through generations to understand their stability and evolutionary implications. Biomarker detection techniques also pave the way to identify molecular markers for desirable phenotypes in conservation, aquaculture, and supplementation. This research could be applied to understand the molecular basis of differences in thermal resilience among populations of westslope cutthroat trout and other salmonids.

How does multi-omic variation contribute to population differentiation, genetic variation, and adaptive potential in natural systems? My research group will study multi-omic variation to better understand how populations adapt to different environments. How do populations and species cope with similar environmental stressors across broad geographic ranges? Are the same molecular mechanisms (e.g., genetic vs. epigenetic variation) and loci involved regardless of population, or do populations find different molecular pathways to the same physiological or phenotypic endpoint? What implications would different pathways have for adaptive potential in changing climates (e.g., purely genetic vs. purely plastic phenotypes) and the speed of adaptation? Over time, epigenetic variation can also lead to point mutations and altered genome structure due to control over transposable element (jumping gene) movement. Could epigenetic variation contribute to genetic divergence among populations and accelerate evolutionary dynamics? A multi-species approach to this research would shed light on how multi-omic variation contributes to adaptation and divergence in natural systems.

Do populations differ in their capacity for plastic responses to environmental change? Both standing genetic variation and capacity for plasticity could help organisms cope with climate change. Capacity for plasticity can evolve, with some populations having greater capacity for plasticity than others. This can have considerable implications for a population's potential to respond to contemporary stressors such as climate change, urbanization, and pollutants. Population-level assessment of these sources of variation will be crucial to understand whether salmonids can cope with climate change and other anthropogenic effects. This research could also find molecular markers linked to colonization and competitive success in salmonids such as Chinook and Atlantic salmon in the Great Lakes, and Atlantic and pink salmon in the Canadian Arctic.

(1) Venney et al. 2016 *Mol. Ecol.* **(2)** Venney et al. 2020 *Mol. Biol. Evol.* **(3)** Venney et al. 2021 *Heredity (Edinb).* **(4)** Venney et al. 2021 *Ecol. Evol.* **(5)** Venney et al. 2023 *Mol. Ecol. Resour*. **(6)** Venney et al. 2022 *Proc. R. Soc. B: Biol.* **(7)** Venney et al. 2023 *Evolution.*

Teaching Statement

I never thought I would become a scientist until I discovered my passion for science during my undergraduate studies. This was partially due to a lack of understanding of what science was – not memorization and repetition, but the underlying curiosity, critical thinking, trial and error, and eventual understanding of unsolved questions that makes science come alive for students. My main goals in teaching are thus to further the understanding and engagement of students who are already excited about biology and evolution, and to ignite or rekindle the same interest in students who may need additional motivation. My teaching approach aims to foster critical thinking and discussion in a welcoming and inclusive environment, leading to a deeper understanding of sometimes complex scientific concepts. Thanks to my previous undergraduate teaching experience, I excel at making complex topics understandable to students. Lectures on difficult topics in genetics, evolution, and molecular biology using plain language, clear examples, and in-class activities will simplify these topics for students and improve their learning experience. My proudest teaching moments were not due to student grades, but rather seeing improved understanding, enthusiasm, and initiative from my students over the course of a semester. I will work to create a safe environment for students to ask questions, discuss ideas, be wrong, and learn from their misconceptions while receiving frequent feedback and support. I am the most effective as an educator when I can discuss with students rather than unilaterally delivering course content. I aim to help the next generation of scientists develop the same excitement for scientific research that I found in my undergraduate studies while also developing critical thinking and other skills that will be useful for students regardless of their career path.

Due to my research bridging a gap between evolution, ecology and molecular biology, I could teach courses in both the Ecology, Evolution, and Environmental Biology and the Molecular, Cellular and Developmental Biology undergraduate programs.

Through my teaching opportunities during graduate school, I have been exposed to various teaching philosophies ranging from traditional lecture-based courses to reverse classrooms where students cover course content on their own time and complete assignments during each lecture to reinforce their learning. I have assisted with courses including Genetics, Evolution, and Introductory Molecular Biology and served as a co-instructor for a course on Evolutionary, Ecological, and Environmental Genetics where I created course content. These courses have ranged from small classes of ~20 students where I worked with each student daily to large courses with more than 300 students and minimal discussion. I particularly enjoyed running Genetics labs as I was able to discuss my favourite subject one-on-one with each student weekly. I aimed to share my passion for genetics despite it being one of the most difficult and dreaded courses in biology. Many of my students joined Daniel Heath's lab as undergraduate volunteers and graduate students which speaks to my ability to make a complex topic like genetics enjoyable and understandable in a way that prompts further engagement. Based on my previous teaching experience and feedback from students on various methods, I would employ a mixed strategy to teaching to ensure that students benefit from individual attention during the course while catering to different learning styles.

I will supplement lectures with activities that provide regular opportunities for discussion with myself, teaching assistants, and student peers. One of my favourite teaching memories came from a small special topics course on Evolutionary, Ecological, and Environmental Genetics. We mentored the class through an experiment assessing the effects of water flow on microbial communities in Detroit River and Lake Erie water bodies. The experiment was entirely conceived, designed, and performed by students and led to a draft manuscript which was submitted for publication. This experience gave students a crash course on the scientific method and brought the classroom to life in ways that lectures and textbooks could not. While this exercise would not be possible with a large class, I will ensure that students would have opportunities for discussion and feedback regardless of class size through different activities. This could include reading and discussing relevant journal articles as a group, friendly debates, group discussions on contemporary scientific topics, poster sessions, class-led research experiments, and more. These exercises would increase student engagement due to interactions with myself and my teaching assistants. They would also allow me to gauge how well students understand the course content so that I can modify or supplement my lectures as needed. While these approaches may be more difficult or infrequent in large classes, I will find other methods to gauge student progress, such as mini quizzes at the beginning of class or online anonymous feedback forms. I see teaching as a mutualistic process through which both the students and I should learn and improve throughout the course. I also realize that every student's needs and strengths are different. Students come from all backgrounds and all walks of life. A one size fits all approach to teaching will only benefit those that are suited to that approach. Fostering one-on-one interactions with students through discussion activities will allow me to assess each student's individual comfort and skill level in the course and meld my teaching strategy in response. Opportunities for teamwork and peer feedback will help students build interpersonal skills which will be helpful regardless of their career path. Deviating from the traditional format of lectures and exams will help students develop critical thinking skills and a deeper understanding of the course material.

As with my undergraduate teaching strategy, I take a highly individualistic approach to graduate and undergraduate research mentorship. I have had the opportunity to mentor [redacted]. These students came from diverse backgrounds, had unique strengths, and required guidance and support in different areas. My approach to mentorship will be different with each student and adapt to their individual needs and method of learning rather than using a one-size-fits-all approach, similar to my approach for undergraduate teaching. Some of the students I have mentored required little guidance and only occasional assistance as they progressed in their projects. Others required attention daily or several times a week until they felt confident enough to independently pursue their research goals. I cultivate healthy working relationships with my mentees, allowing for open communication and a judgement free research environment that fosters individual growth and independence in research. I will endeavour to support my mentees by providing them with opportunities and support to improve their skills in areas that may be lacking. This could entail receiving additional attention from myself, funding additional training through workshops or courses, or building collaborations with researchers in academia, government, and elsewhere. Setbacks, unforeseen roadblocks, and rejection are all part of the scientific process, and I ensure that my mentees understand this and find camaraderie in myself and their peers. Some students need more support than others and therefore I tailor my mentorship approach to each individual student. My ultimate goal is to help shape each student into an independent, inquisitive researcher who can solve problems and think critically about their research questions.

I also believe in promoting healthy work attitudes and work-life balance for graduate students after seeing the negative effects of succumbing to overwork culture in graduate school. While I will still have expectations for student productivity, I do not believe that productivity should come at the cost of mental health. Productivity in my lab will not solely be measured by publication success and other tangible research outcomes, but also by individual growth, skill development, engagement with research, and collaborative efforts. I have been fortunate to have supportive, understanding supervisors for my graduate and postdoctoral studies and have seen firsthand that cultivating a positive, supportive research environment leads to a productive and collaborative research team. This also allows me to modify my approach to ensure mentees from underrepresented groups have the opportunities, flexibility, and support they need to excel in a safe, inclusive work environment.

Commitment to Equity, Diversity, and Inclusion

One of the main deficiencies in academia is the unconscious bias against and exclusion of underrepresented groups. I have experienced this firsthand as a woman in science, watching my female and minority peers slowly leave academia due to both visible and invisible barriers. However, there are many other initial barriers to underrepresented groups entering academia: lack of interest in science due to insufficient outreach, inaccessibility of teaching styles or materials for students from diverse educational backgrounds or learning styles, limited opportunities and funding for research, and inhospitable or unsafe research environments. I believe that combatting these barriers to underrepresented groups will be a career-long challenge requiring me to continually assess and reassess my own approaches and implicit biases in teaching and mentorship. New students will bring new instances of intersectionality which will require my EDI actions to evolve over time.

For research, EDI efforts begin in the lab. Fostering an inclusive, diverse research environment will bring new ideas, approaches, and knowledge that would be lost without the engagement of underrepresented groups. I will build a research group based on mutual respect and consideration, recruiting students through networking at conferences and on social media. We will hold regular lab meetings with the expectation that my students will attend, provide feedback, and collaborate within and outside my laboratory. This will ensure consistent oversight and feedback for students and ensure that I am meeting students' individual needs to succeed and thrive in their research. With continual feedback from lab members, we will build a dynamic code of conduct and guide for best practices to ensure the lab remains safe and inclusive. I will also be flexible based on each student's needs with respect to language skills, childcare/family care, and other needs to increase inclusiveness based on each student's circumstances and accessibility requirements.

Conducting research in your non-native language is particularly costly in terms of time – something that I have learned as an anglophone postdoctoral fellow at a francophone university. Reading papers, writing manuscripts, and preparing presentations in your non-native language requires a substantial time investment compared to working in your native language. This often leads to students becoming discouraged from attending conferences, sharing their research, and ultimately publishing their work if they do not feel comfortable in English. As a postdoctoral fellow, I have made myself available as an English editor for my colleagues and have been fortunate that they have returned the favour by editing my French. I will continue this practice as a professor and create an expectation that lab members will help one another with language editing. I will also fund additional English writing training for interested students to reduce this barrier and help them flourish as researchers. Even though science is mainly disseminated in English, many scientifically valuable studies are published in other languages. This creates pockets of research that follow national or linguistic divides, leading to rifts in the literature that can further marginalize researchers from underrepresented minorities since their work cannot be appreciated by the wider anglophone scientific community (and vice versa). This also creates barriers for science communication in local communities near our research sites, preventing their involvement in science and further investment in the conservation of their local ecosystems. This is particularly an issue in Canadian communities with few anglophones, such as Indigenous and French-Canadian communities. While these issues are systemic with science and society converging towards English, I will endeavour to disseminate my work in the relevant local language(s), either through my skills in written and spoken French, or through the use of interpreters. I will endeavour to involve Indigenous communities and groups in my research and incorporate traditional knowledge in the study and conservation of species of shared interest. I will also encourage my students to share their research in any language they feel comfortable with to reduce linguistic and geographic barriers to science communication and collaboration.

Financial barriers also prevent students from pursuing research opportunities. The cost of living is rapidly increasing in Canada and it is becoming financially unfeasible for many students to pursue graduate studies. Most graduate students currently live under the poverty line in Canada which creates an additional entry barrier for students from low income and monoparental homes without familial wealth. These students must often work a second job to afford the necessities of life which detracts from their commitment to their research and increases the chances of burnout. This issue is exacerbated for international students who pay increased tuition, another barrier preventing prospective students from underrepresented groups from pursuing graduate studies. I was a main organizer of the Quebec City Support Our Science walkout in May 2023, a national movement advocating for increased federal funding for graduate students and postdoctoral researchers. Our ongoing request is for the federal government to increase the monetary value and the number of awards offered by the tri-council scholarships. This would enable principal investigators to ask for increased stipends for high quality personnel on federal research grants since reasonable requests for stipends are dictated by the current tri-council award amounts. I will continue to advocate for increased stipends for high quality personnel in my own research group, speaking from a position of power as a professor. I will endeavour to pay my students a livable wage and provide adequate funding for all research opportunities. Additionally, the initial costs of conferences and fieldwork can be a burden on students who must take on debt and wait for the university to reimburse them. I will do everything in my power to prevent students from having to shoulder that debt themselves (e.g., through a lab credit card, if possible). I also hope to provide paid undergraduate research positions so that students interested in research can explore these opportunities without worrying about finances, which is a major barrier to undergraduate student engagement in research labs. These actions will ideally reduce the initial financial barriers to pursuing graduate studies.

I will use a mixed method of teaching at the undergraduate level, incorporating both lectures and discussion activities as outlined in my Teaching Statement. The use of discussion-based assignments will allow for one-on-one communication with students, creating a safe environment to identify the individual needs of students. The use of different learning methods and assignments will cater to students with diverse learning styles. I will also implement a code of conduct in courses similar to what I develop in my research group. This code of conduct will evolve over time with input from students. With these approaches, I aim to ensure that all individuals, especially those from underrepresented groups, feel supported and safe in my classes while acknowledging that a one-size-fits-all approach to teaching does not work. Every student I meet will have their own story and background which I will need to adapt to where possible. These diverse stories are currently lacking in academia. One of the first chronological steps to increase diversity in academia is to ensure that underrepresented groups are welcomed and supported during their undergraduate studies.

I will also engage in public outreach programs to motivate undergraduate and high school students and community members to become involved in scientific research in an accessible manner. I participated in community science and outreach events such as the University of Windsor's Science Academy for high school students and a community water sampling effort. These free activities provided a unique opportunity for students and community members to engage in science with no financial barrier limiting who can become involved. Organizing similar activities as a professor would increase interest in scientific education and career opportunities for students who have never considered science as a career path.

Overall, EDI considerations will be a lifelong learning experience and engagement requiring consistent effort and reevaluation, both by myself and by the institution. However, reducing or removing barriers for underrepresented groups and maintaining a safe, inclusive learning environment is an important first step in my efforts to combat underrepresentation in the university and academia.

Five Most Impactful Publications

*** Equal contribution is denoted by an asterisk.**

1. **Anastasiadi D*, Venney CJ***, Bernatchez L, Wellenreuther M. (2021) Epigenetic inheritance and reproductive mode in plants and animals. *Trends in Ecology and Evolution* 36(12): 1124-1140. **(IF=20.6, 57 citations)**

   We performed a systematic review of over 550 papers (of >2200 screened papers) on epigenetic inheritance, identifying trends in the frequency of epigenetic inheritance based on reproductive mode, and discussing novel questions on its evolutionary implications.

2. **Venney CJ*, Cayuela H*,** Rougeux C, Laporte M, Mérot C, Normandeau E, Leitwein M, Dorant Y, Præbel K, Kenchington E, Clément M, Sirois P, Bernatchez L. (2022) Genome-wide DNA methylation predicts environmentally-driven life history variation in a marine fish. *Evolution* 77(1): 186-198. **(IF=3.3, 1 citation)**

   We combined whole genome and methylome data to determine what drives beach-spawning vs. demersal life history tactics in the capelin fish, determining that there is little genetic differentiation but considerable epigenetic divergence between tactics which is partially genetically controlled. While this paper came out very recently in 2022, it is one of the first studies linking both epigenetic and genetic variation to life history tactics.

3. **Venney CJ**, Wellband KW, Normandeau E, Houle C, Garant D, Audet C, Bernatchez L. (2022) Thermal regime during parental sexual maturation, but not during offspring rearing, modulates DNA methylation in brook charr (*Salvelinus fontinalis*). *Proceedings of the Royal Society B: Biological Sciences* 289(1974): 20220670. **(IF=5.5, 10 citations)**

   We determined that offspring methylation state is primarily influenced by parental temperature during sexual maturation rather than the offspring’s own environment, with important implications for brook charr supplementation efforts during climate change.

4. **Venney CJ**, Bouchard R, April J, Normandeau E, Lecomte L, Côté G, Bernatchez L. (2023) Captive rearing effects on the methylome of Atlantic salmon after oceanic migration: sex-specificity and intergenerational stability. *Molecular Ecology Resources*. **(IF=8.7, 1 citation)**

   We showed that methylation changes induced by captive rearing in Atlantic salmon persist after oceanic migration with little transmission to offspring in a natural environment. However, the marks that were inherited showed nonadditive, unpredictable patterns of inheritance which may be difficult to account for in conservation efforts.

5. **Venney CJ**, Wellband KW, Heath DD. (2021) Rearing environment affects the genetic architecture and plasticity of DNA methylation in Chinook salmon. *Heredity* 126: 38-49. **(IF=3.8, 26 citations)**

   We used candidate gene methylation analysis and factorial breeding experiments to determine how the underlying genetic architecture of DNA methylation is affected by hatchery vs. semi-natural rearing, finding greater maternal effects in the semi-natural environment but greater additive (heritable) and nonadditive (dominance or epistasis) effects in the hatchery environment.